

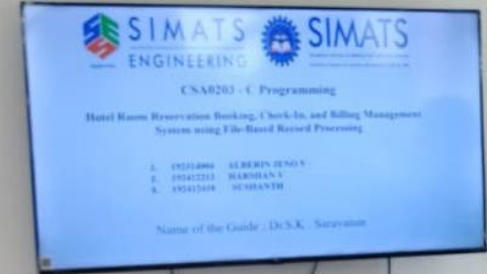
GPS Map Camera

Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026444° Long 80.017029°
Tuesday, 25/08/2026 02:08 PM GMT +05:30

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CSA0203 - C Programming

Hotel Room Reservation Booking, Check-In, and Billing Management System using File-Based Record Processing

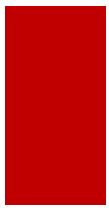
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Name of the Guide : Dr.S.K . Saravanan



INTRODUCTION

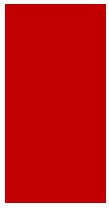
- ✓ Hotels require an efficient system to manage room reservations, guest check-in/check-out, and billing.
- ✓ Traditional manual methods are time-consuming, prone to errors, and difficult to maintain.
- ✓ This project is developed using the **C programming language** with **file handling** to store guest and booking records permanently.
- ✓ The system automates room booking, availability tracking, guest management, and bill generation through a menu-driven interface.
- ✓ It improves accuracy, reduces manual effort, and demonstrates the practical use of **structures, functions, and file-based record processing** in C programming.



PROBLEM STATEMENT



- ✓ Manual hotel management is slow and error-prone.
- ✓ Room booking and availability tracking are difficult.
- ✓ Guest records are hard to store and retrieve.
- ✓ Billing calculations may contain errors.
- ✓ A file-based C application is needed to automate reservations, check-in/check-out, and billing while maintaining permanent records.



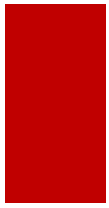
PROPOSED SOLUTION



The proposed system is a menu-driven application developed in C programming that automates hotel operations using file-based record processing. It manages room reservations, guest check-in/check-out, and billing while securely storing records for future access.

Key Features

- Room reservation and availability management
- Guest check-in and check-out
- Automatic billing and payment calculation
- File-based storage for permanent records
- Search, update, and delete guest information
- User-friendly menu-driven interface

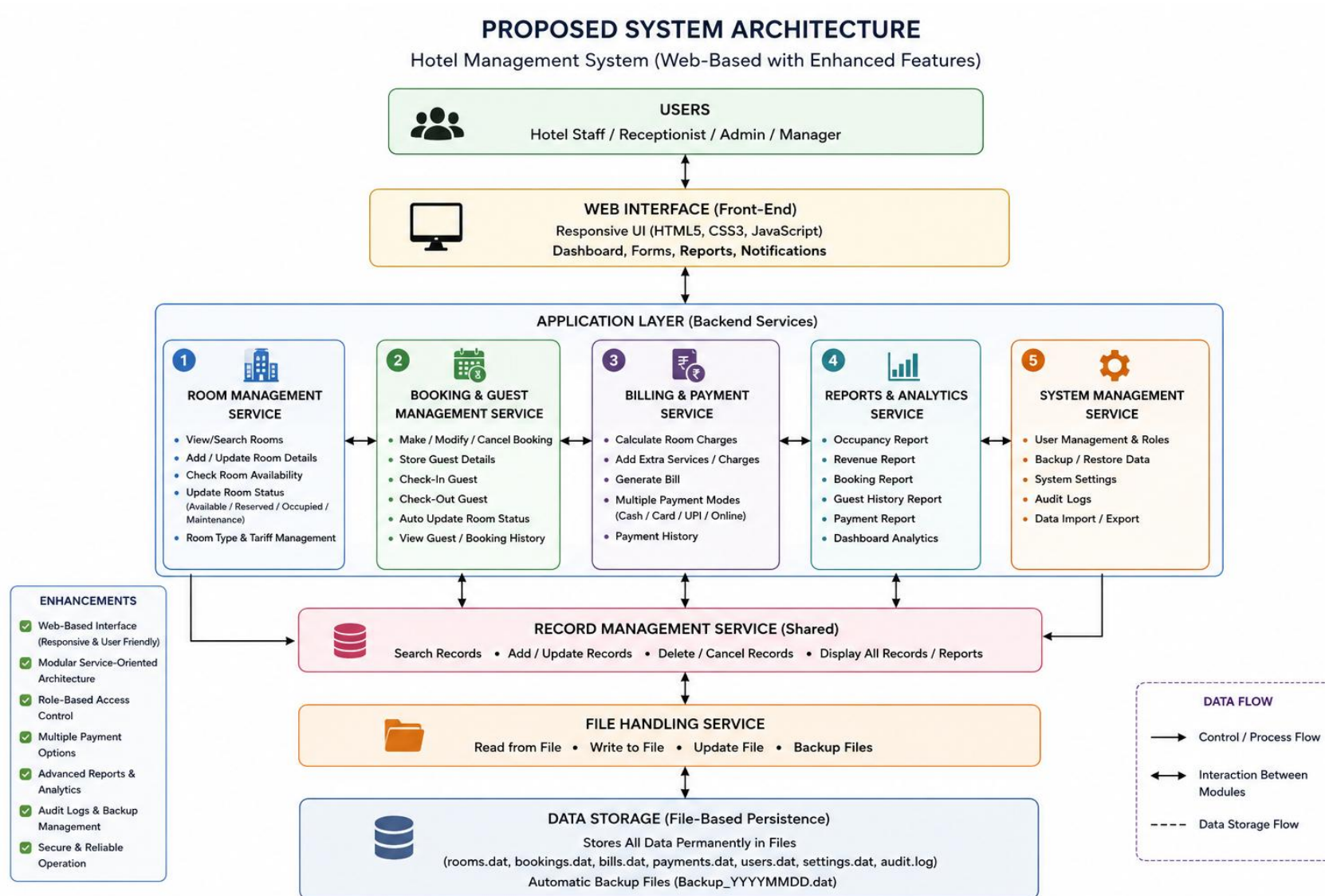


Functionalities



- **Room Information & Availability** View available and occupied rooms.
- Update room status after booking or check-out.
- **Booking Management** Reserve rooms for guests.
- Store guest details using file handling.
- **Check-In & Check-Out** Register guest check-in.
- Process check-out and update room availability.
- **Billing & Payment** Calculate room charges automatically.
- Generate and display the final bill.
- **Record Management** Search, update, delete, and display guest records.
- **File-Based Storage** Save and retrieve records permanently using files.

SYSTEM ARCHITECTURE





Module 1: Room & Reservation Management



- **Room Management:** Maintains complete information about all hotel rooms, including room number, room type, price, and availability.
- **Availability Checking:** Checks the current status of rooms and displays only the rooms that are available for booking.
- **Customer Reservation:** Allows staff to enter customer details and reserve a suitable room based on availability.
- **Booking Validation:** Verifies the booking details and prevents duplicate bookings or reservation of already occupied rooms.
- **Record Storage:** Stores room and reservation details securely in **binary files**, so the information is available even after restarting the system.
- **Room Status Tracking:** Automatically updates the room status throughout the process: **Available → Reserved → Occupied → Available** after checkout.



Module 2: Check-In, Check-Out & Billing Management

- **Customer Check-In:** Allows the staff to confirm the reservation and check the customer into the assigned room.
- **Occupancy Management:** Changes the room status from **Reserved** → **Occupied** after successful check-in.
- **Check-Out Processing:** Records the customer's checkout and verifies the stay details before releasing the room.
- **Bill Calculation:** Calculates the total amount based on the **room rate and number of days stayed**.
- **Bill Generation & Storage:** Generates the final bill with customer, room, stay, and payment details and stores it permanently.
- **Room Status Update:** After checkout and bill generation, the room is automatically changed from **Occupied** → **Available**, making it ready for the next customer.



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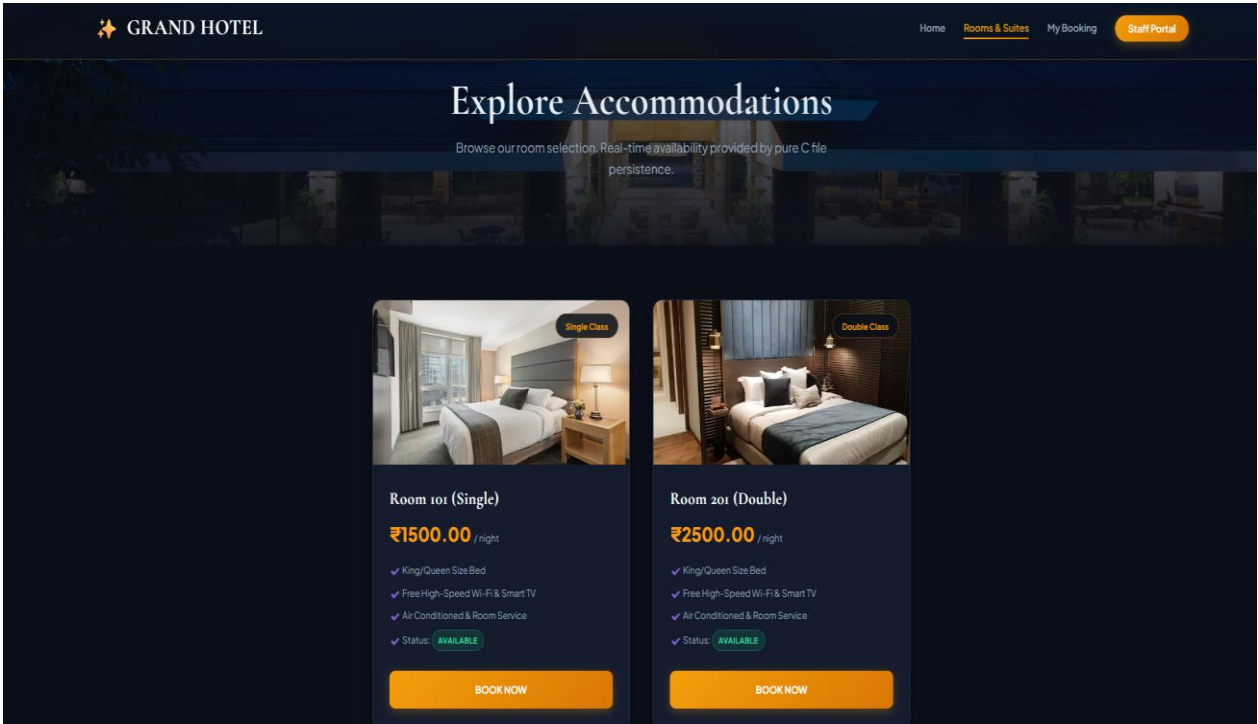
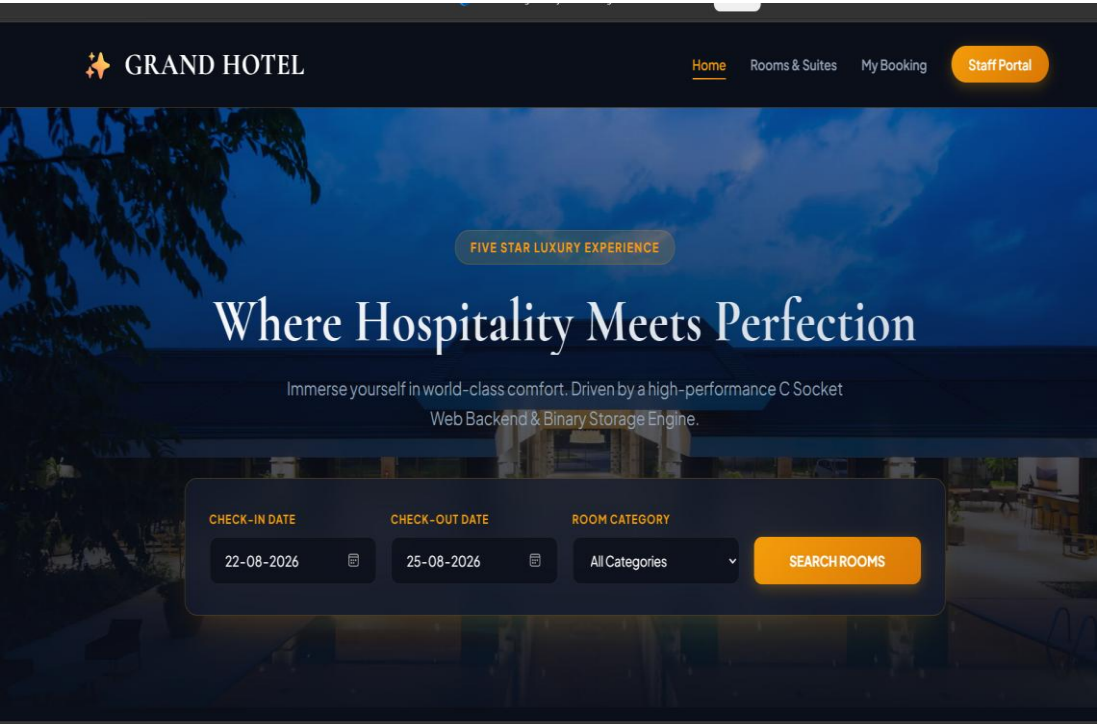
Module 3: File Management & System Control



- **File-Based Data Storage:** Stores all important hotel information using binary files such as `rooms.dat`, `bookings.dat`, and `bills.dat`.
- **Data Persistence:** Keeps room, booking, and billing records even after the server or system is restarted.
- **Record Management:** Allows the system to read, update, and maintain existing hotel records whenever required.
- **Data Consistency:** Ensures that changes in room status, reservations, and billing records are properly synchronized.
- **Automatic Data Initialization:** If the required data files are not available, the system creates them and initializes the default room records.
- **Reliable System Operation:** Provides continuous management of hotel data while reducing manual record keeping and improving accuracy.



Output





Output



 GRAND HOTEL

HomeRooms & SuitesMy BookingStaff Portal



Room Reservation Form

Fill in your details below to secure your luxury stay.

SELECT ROOM

Room 101 (Single) - ₹1500.00/night

FULL NAME

Enter guest full name

PHONE NUMBER

e.g. 9876543210

EMAIL ADDRESS

e.g. guest@example.com

CHECK-IN DATE

22-08-2026

CHECK-OUT DATE

25-08-2026

NUMBER OF GUESTS

1

Estimated Stay Duration:3 nights

Estimated Room Charge:₹4500.00

CONFIRM RESERVATION



Output



 **GRAND HOTEL** [ADMIN]

[Home](#) [Rooms](#) [My Booking](#) [Staff Portal](#) [Logout \(admin\)](#)

Management Dashboard

Welcome, admin (admin) | [Logout](#)

TOTAL ROOMS
5

AVAILABLE
4

RESERVED
1

OCCUPIED
0

OCCUPANCY RATE
0%

TOTAL REVENUE
₹9900.00

+ Add Room

% Room Maintenance

🔑 Check-In

📄 Generate Bill

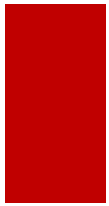
💵 Record Payment

🚪 Check-Out

👤 Add User

Room Inventory

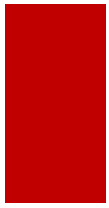
ROOM NO	TYPE	RATE / NIGHT	STATUS
Room 101	Single	₹1500.00	AVAILABLE
Room 102	Single	₹1500.00	RESERVED
Room 201	Double	₹2500.00	AVAILABLE
Room 202	Double	₹2500.00	AVAILABLE



CONCLUSION



The Hotel Room Reservation Booking, Check-In, and Billing Management System demonstrates how file-based record processing can be used to automate essential hotel operations using the C programming language. The system efficiently manages room reservations, guest check-in and check-out, billing, and record maintenance through a simple menu-driven interface. By utilizing structures, functions, and file handling, it ensures accurate data management, reduces manual errors, and provides permanent storage of records. This project serves as a practical application of fundamental C programming concepts and can be further enhanced with features such as user authentication, database integration, and an online reservation interface.



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THANK YOU